Combinatorial Proof and Bijection Principle

- **Definition**
  A **combinatorial proof** of an identity is a proof that uses counting arguments to prove that both sides of the identity count the same objects but in different ways.

- **Bijection principle (type of combinatorial proof)**
  Two sets U and V have the same size, i.e., $|U| = |V|$, if and only if there is a bijection function f from U to V (i.e., one-to-one and onto)